Arcs complete outside a prescribed conic

Exact defect, equality rigidity, and $\rho_{\mathcal{C}}(16) = 9$

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Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$. We study arcs $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ whose secants cover every point outside $A \cup \mathcal{C}$, and write $\rho_{\mathcal{C}}(q)$ for their minimum size. Splitting the classical first two secant-index equations over an arbitrary prescribed hole and its complement gives an exact defect identity whose remainder is a sum of nonnegative local terms. For a conic this yields an exact integer lower bound $L_2(q)$, and in particular

$$\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.$$

Vanishing defect turns the concurrence points of disjoint secants into a matching design, giving equality rigidity and quantitative stability; for ten points equality occurs only in $\mathrm{PG}(2,8)$, and the concurrence design is the regular-hyperoval design. Projective averaging transfers ordinary complete arcs to upper bounds in the relative problem. Explicit witnesses attain $L_2(q)$ for $q = 5, 8, 9, 11$. Finally, a kernel-checked exhaustive covering certificate excludes every eight-arc over $\mathbb{F}_{16}$: its ordinary uncovered locus cannot lie on a quadratic avoiding the arc. A certificate-checked nine-point witness therefore gives $\rho_{\mathcal{C}}(16) = 9$.

1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. An arc is complete when it cannot be enlarged while remaining an arc; equivalently, every point outside it lies on a secant determined by two of its points. Complete arcs and the closely related problem of finding small saturating sets have a substantial literature; see, for example, Ball [5], Kim–Vu [13], and Nagy [16].

Fix a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2,q)$. We impose the arc condition but allow the points of $\mathcal{C}$ to remain uncovered.

Definition 1.1. An arc $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ is $\mathcal{C}$-*complete* if every point of

$$\mathrm{PG}(2,q) \setminus (A \cup \mathcal{C})$$

lies on a secant of A. Define

$$\rho_{\mathcal{C}}(q) = \min\{|A| : A \text{ is a } \mathcal{C}\text{-complete arc}\}.$$

More generally, if $\mathcal{H}$ is any set disjoint from an arc A, write

$$\mathcal{U}_{\mathcal{H}}(A) = \mathrm{PG}(2,q) \setminus \left(A \cup \mathcal{H} \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right).$$

We abbreviate $U(A) = \mathcal{U}_{\varnothing}(A)$ for the ordinary-uncovered locus and retain $\mathcal{U}_{\mathcal{C}}(A)$ for the conic-relative locus. Thus A is $\mathcal{C}$-complete exactly when $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$.

The minimum is attained because the finite point set $\mathrm{PG}(2,q)\setminus\mathcal{C}$ contains an inclusion-maximal arc A. For every point $x\notin A\cup\mathcal{C}$, maximality forces $A\cup\{x\}$ to contain three collinear points, so x lies on a secant of A. Thus A is $\mathcal{C}$-complete. All nonsingular conics in $\mathrm{PG}(2,q)$ are projectively equivalent, so the numerical value of $\rho_{\mathcal{C}}(q)$ does not depend on the chosen conic.

The following statement collects the main conclusions. For a point x outside a k-arc A, let $r(x)$ be the number of secants of A through x, and put $m=\lfloor k/2\rfloor$ and $N=\binom{k}{2}$. If a set $\mathcal{H}$ disjoint from A is allowed to remain uncovered, let $\mathcal{X}_{\mathcal{H}}(A)$ be the covered points outside $A\cup\mathcal{H}$ and let $I_{\mathcal{H}}(A)=\sum_{y\in\mathcal{H}} r(y)$.

Theorem 1.2 (Defect, lower bound, and $q=16$)**.** *For every arc A and prescribed hole $\mathcal{H}$, define*

$$\Delta_{\mathcal{H}}(A)=N(q-1)-\frac{6}{m}\binom{k}{4}-\frac{I_{\mathcal{H}}(A)}{m}-|\mathcal{X}_{\mathcal{H}}(A)|.$$

Then

$$m\Delta_{\mathcal{H}}(A)=\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)}(r(x)-1)(m-r(x))+\sum_{y\in\mathcal{H}} r(y)(m-r(y)).$$

In particular, if

$$L_2(q)=\min\left\{k\geq 3: q^2-k\leq\binom{k}{2}(q-1)-\frac{6}{\lfloor k/2\rfloor}\binom{k}{4}\right\},$$

then every nonsingular conic satisfies

$$\rho_{\mathcal{C}}(q)\geq L_2(q)\geq\sqrt{2q}+\frac{3}{2}-\frac{8}{\sqrt{2q}}.$$

Moreover, $\rho_{\mathcal{C}}(16)=9$.

The identity is the conceptual step. Its summands record exactly where a secant-index distribution falls short of the extremal pattern. Their vanishing produces the matching design used for equality rigidity, while their total gives stability. The last assertion of Theorem 1.2 has a separate finite input: a checked augmentation certificate reduces all eight-arcs over $\mathbb{F}_{16}$ to 2633 leaves, and checked quadratic-evaluation records reject every leaf.

The surrounding notions differ at the defining containment. Saturating sets need not be arcs, while almost-complete conic subsets select points on the conic [8]. A line hole recovers affine completeness, including the related hyperfocused constructions [17, 10, 14]; see Corollary 3.7. Complete exterior sets point in the opposite direction: their joins miss the conic [9, p. 143]. Thus complete exteriority gives $\mathcal{C}(\mathbb{F}_q)\subseteq U(A)$, whereas $\mathcal{C}$-completeness is equivalent to $U(A)\subseteq\mathcal{C}(\mathbb{F}_q)$.

The point-index equations themselves are classical; compare Hirschfeld [12, Chapter 9], Ball [5], and the recent explicit complete-arc bound of Alabdullah–Hirschfeld [2]. The new step is the exact local remainder after splitting those equations over a prescribed hole and its complement.

The equality structure belongs to the established theory of matching designs. Alspach and Heinrich [4] call a family of s-edge matchings in K_n a $\mathrm{MATCH}(n,s,\lambda)$ design when every pair of independent edges occurs in exactly λ members. Our contribution at this interface is the geometric implication: vanishing of the prescribed-hole defect canonically produces a simple $\mathrm{MATCH}(k,\lfloor k/2\rfloor,1)$ design, represented by concurrence points of secants in one projective plane. The local remainder also measures the graph edges that fail to lie in such maximum-matching cliques.

Sections 2–8 give the main route: exact accounting, equality rigidity, lower and upper bounds, the characteristic-two refinement, and the finite obstruction. The short final section records an elementary inverse consequence: above a sharp incidence threshold, the complete uncovered locus determines its parent arc and automorphism group.

The secant-index equations and the Lunelli–Sce $\sqrt{2q}$ scale are classical. The contributions are the prescribed-hole remainder with its equality and stability consequences, its conic specialization, the uncovered-locus evaluation obstruction, and the certified relative-conic conclusions.

2 The classical secant equations

Throughout the remainder of the paper, $q \geq 3$ and $k \geq 3$. Let Π be a projective plane of order q, and let A be a k-arc in Π. Write $r_A(x)$ for the number of secants of A through $x \in \Pi \setminus A$. We usually write $r(x)$. Put

$$N = \binom{k}{2}, \qquad m = \left\lfloor \frac{k}{2} \right\rfloor.$$

Lemma 2.1. *For every $x \notin A$, one has $r(x) \leq m$.*

Proof. Distinct secants through x cannot share an endpoint in A. Indeed, two lines through the same pair x, a coincide. Hence the secants through x use pairwise disjoint pairs of points of A, of which there are at most $\lfloor k/2 \rfloor$. □

Proposition 2.2 (Classical first and second index equations)**.** *For every k-arc A in a projective plane of order q,*

$$\sum_{x \notin A} r(x) = \binom{k}{2}(q-1), \tag{1}$$

$$\sum_{x \notin A} \binom{r(x)}{2} = 3\binom{k}{4}. \tag{2}$$

Proof. There are $\binom{k}{2}$ secants. Each contains $q+1$ points, exactly two of them in A, and therefore contributes $q-1$ incidences with points outside A. This proves (1).

For (2), count unordered pairs of distinct secants through a common external point. By Lemma 2.1, their four endpoints in A are distinct. Conversely, a four-subset of A has three partitions into two unordered pairs. Each partition gives two secants, which meet in a unique point because the plane is projective. Their intersection is outside A: otherwise one of the two secants would contain three points of the arc. Thus every four-subset contributes exactly three. □

Only the projective-plane axioms and the common line size $q+1$ enter these proofs. In particular, the second equation need not hold in a general finite linear space where two lines may be disjoint.

3 An exact defect identity for prescribed holes

The next theorem is stated for an arbitrary exceptional point set. The conic enters only in later specializations.

Let $\mathcal{H} \subseteq \Pi \setminus A$ be any set of points allowed to remain uncovered. The notation used in the identity is summarized by

$$\begin{array}{r|l} V_{\mathcal{H}}(A) = \Pi \setminus (A \cup \mathcal{H}) & \text{required locus} \\ \mathcal{X}_{\mathcal{H}}(A) = \{x \in V_{\mathcal{H}}(A) : r(x) > 0\} & \text{covered required points} \\ \mathcal{U}_{\mathcal{H}}(A) = V_{\mathcal{H}}(A) \setminus \mathcal{X}_{\mathcal{H}}(A) & \text{uncovered required points} \\ I_{\mathcal{H}}(A) = \sum_{y \in \mathcal{H}} r(y) & \text{secant incidence spent on the holes.} \end{array}$$

The third line agrees with the uncovered-locus notation introduced in Definition 1.1; in particular, $U = \mathcal{U}_{\varnothing}$ and the conic-relative locus is $\mathcal{U}_{\mathcal{C}}$.

Splitting Proposition 2.2 over $V_{\mathcal{H}}(A)$ and $\mathcal{H}$ gives

$$\sum_{x \in V_{\mathcal{H}}(A)} r(x) = N(q-1) - I_{\mathcal{H}}(A), \tag{3}$$

$$\sum_{x \in V_{\mathcal{H}}(A)} \binom{r(x)}{2} + \sum_{y \in \mathcal{H}} \binom{r(y)}{2} = 3\binom{k}{4}. \tag{4}$$

Theorem 3.1 (Prescribed-hole defect identity)**.** *Let A be a k-arc in a projective plane of order q, let $\mathcal{H} \subseteq \Pi \setminus A$, and put $m = \lfloor k/2 \rfloor$. Define*

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

It is the exact coverage capacity left after the universal second-moment and hole-incidence losses; the identity below resolves that capacity into local nonextremal-index contributions. Then

$$\boxed{m\Delta_{\mathcal{H}}(A) = \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y \in \mathcal{H}} r(y)(m-r(y)).} \tag{5}$$

In particular, $\Delta_{\mathcal{H}}(A) \geq 0$.

Proof. Only points of $\mathcal{X}_{\mathcal{H}}(A)$ contribute to the sum in (3). Multiplying the definition of $\Delta_{\mathcal{H}}(A)$ by m and using (3) gives

$$m\Delta_{\mathcal{H}}(A) = m \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} r(x) + (m-1)I_{\mathcal{H}}(A) - m|\mathcal{X}_{\mathcal{H}}(A)| - 6\binom{k}{4}.$$

By (4),

$$6\binom{k}{4} = 2\sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} \binom{r(x)}{2} + 2\sum_{y \in \mathcal{H}} \binom{r(y)}{2}.$$

Substitution first separates the required-point and hole terms:

$$\begin{aligned} m\Delta_{\mathcal{H}}(A) = &\sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} \left(m(r(x)-1) - 2\binom{r(x)}{2} \right) \\ &+ \sum_{y \in \mathcal{H}} \left((m-1)r(y) - 2\binom{r(y)}{2} \right). \end{aligned}$$

The pointwise identities

$$m(r-1) - 2\binom{r}{2} = (r-1)(m-r),$$

$$(m-1)r - 2\binom{r}{2} = r(m-r),$$

now give (5). Every term is nonnegative by Lemma 2.1. $\square$

Corollary 3.2 (Coverage and uncovered-locus bounds)**.** *Under the hypotheses of Theorem 3.1,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \le N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}, \tag{6}$$

and

$$|\mathcal{U}_{\mathcal{H}}(A)| \ge |V_{\mathcal{H}}(A)| - N(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{H}}(A)}{m}. \tag{7}$$

Equality holds in either inequality if and only if

$$r(x) \in \{1, m\} \quad (x \in \mathcal{X}_{\mathcal{H}}(A)), \qquad r(y) \in \{0, m\} \quad (y \in \mathcal{H}).$$

Proof. The two inequalities are respectively $\Delta_{\mathcal{H}}(A) \ge 0$ and its rearrangement using $|\mathcal{U}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| - |\mathcal{X}_{\mathcal{H}}(A)|$. The equality statement follows from the vanishing of every nonnegative term in (5). □

For later use, we record the complete-outside consequence before imposing any geometry on the holes.

Corollary 3.3 (Arbitrary prescribed holes)**.** *Let $|\mathcal{H}| = h$. If A is a k-arc disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 + q + 1 - k - h \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}. \tag{8}$$

Proof. Completeness gives $|\mathcal{X}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| = q^2 + q + 1 - k - h$. Substitute this equality into (6). □

The exact remainder also quantifies near equality.

Corollary 3.4 (Quantitative stability)**.** *Assume $m \ge 3$ and put*

$$M = \{x \in \mathcal{X}_{\mathcal{H}}(A) : 2 \le r(x) \le m-1\},$$
$$J = \{y \in \mathcal{H} : 1 \le r(y) \le m-1\}.$$

Then

$$(m-2)|M| + (m-1)|J| \le m\Delta_{\mathcal{H}}(A). \tag{9}$$

Proof. For $2 \le r \le m-1$, $(r-1)(m-r) \ge m-2$. For $1 \le r \le m-1$, $r(m-r) \ge m-1$. Apply these estimates to (5). □

Thus small defect forces almost every multiply covered required point to have the maximum possible multiplicity m, while almost every hole point met by a secant also has multiplicity m. This is the precise stability content behind the equality heuristic.

3.1 Concurrency designs at equality

The equality criterion has a global incidence interpretation. Label the vertices of K_k by the points of A, so that its edges label the secants of A. Two such edges are adjacent in the Kneser graph $KG(k, 2)$ exactly when the corresponding secants have disjoint endpoint pairs.

Theorem 3.5 (Concurrency decomposition and equality rigidity)**.** *Suppose $k \geq 4$. For each $x \notin A$ with $r(x) \geq 2$, the secants through x form an $r(x)$-edge matching M_x in K_k, and*

$$E(KG(k,2)) = \dot{\bigcup_{x \notin A,\ r(x) \geq 2}} E(K_{M_x}).$$

Here K_{M_x} is the clique whose vertices are the edges of M_x.

If $\Delta_{\mathcal{H}}(A) = 0$, every M_x in this decomposition has $m = \lfloor k/2 \rfloor$ edges. Thus the family $(M_x)_{r(x)=m}$ is a simple MATCH$(k, m, 1)$ *design. If $Z = \{x \notin A : r(x) = m\}$, then*

$$|Z| = \frac{3\binom{k}{4}}{\binom{m}{2}} = \begin{cases} (k-1)(k-3), & k \text{ even}, \\ k(k-2), & k \text{ odd}, \end{cases}$$

and every secant of A contains $k-3$ points of Z when k is even and $k-2$ points of Z when k is odd.

Proof. Secants through a point outside an arc have pairwise disjoint endpoint pairs, so they form a matching. Conversely, two secants with disjoint endpoint pairs meet in a unique point outside A. Their edge in $KG(k,2)$ therefore belongs to exactly one concurrence clique, proving the decomposition.

If the defect vanishes, Theorem 3.1 gives $r(x) \in \{1, m\}$ off $\mathcal{H}$ and $r(x) \in \{0, m\}$ on $\mathcal{H}$. Every concurrence point has index at least two, so all the cliques above have order m. The second index equation now gives $|Z|\binom{m}{2} = 3\binom{k}{4}$.

Fix a secant. It is disjoint from $\binom{k-2}{2}$ other secants, and each point of Z on it accounts for $m-1$ of them. Hence it contains $\binom{k-2}{2}/(m-1)$ points of Z, which has the two displayed values. $\square$

After any projective duality, the $\binom{k}{2}$ secants become points. They lie on k star lines of size $k-1$, one for each point of A, and on the $|Z|$ matching lines of size m. Two dual secant-points lie on a star line when their endpoint pairs meet and on a unique matching line when they are disjoint. Thus zero defect does not merely prescribe the indices: it gives a rank-three projective realization of the star–matching pairwise-balanced design on the edges of K_k.

The same decomposition strengthens Corollary 3.4 from a count of nonextremal centres to a count of affected pairs of secants.

Corollary 3.6 (Bad-edge stability)**.** *Let E_{bad} be the set of edges of $KG(k,2)$ lying in a concurrence clique of order $2 \leq r(x) \leq m-1$. Then*

$$|E_{\text{bad}}| \leq \frac{m(m-1)}{2} \Delta_{\mathcal{H}}(A).$$

Proof. At an off-hole centre of index r,

$$\binom{r}{2} \leq \frac{m-1}{2}(r-1)(m-r),$$

For a hole centre whose clique contributes to E_{bad}, the definition gives $2 \leq r \leq m-1$, and its defect weight $r(m-r)$ is at least the off-hole weight $(r-1)(m-r)$. Hole centres of index m are excluded from E_{bad} and contribute nothing to its edge count. Sum the resulting inequalities and apply (5). $\square$

The line-hole case now makes the connection with complete affine arcs explicit.

Corollary 3.7 (Complete affine arcs)**.** *Let L_∞ be a line in a projective plane of order q, and let A be a k-arc disjoint from L_∞. Then A is complete as an arc in the affine plane $\Pi \setminus L_\infty$ if and only if it is complete outside L_∞. In that case*

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}\binom{k}{2}. \tag{10}$$

Equality holds if and only if every affine point outside A has secant index in $\{1, m\}$ and every point of L_∞ has secant index in $\{0, m\}$.

Proof. Maximality in the affine plane is equivalent to every affine point outside A lying on a secant of A, which is exactly completeness outside L_∞. Every secant meets L_∞ once, so $I_{L_\infty}(A) = \binom{k}{2}$. Now apply Corollaries 3.3 and 3.2; the latter also gives the equality criterion. □

4 Specialization to a conic

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$, and let $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ be a k-arc. Since $|\mathcal{C}| = q+1$, there are $q^2 - k$ points outside $A \cup \mathcal{C}$. Write

$$I_{\mathcal{C}}(A) = \sum_{y \in \mathcal{C}} r(y) = \sum_{\ell \text{ an } A\text{-secant}} |\ell \cap \mathcal{C}|.$$

Corollary 4.1 (Conic-loss and uncovered-locus inequalities)**.** *For every such arc,*

$$|\mathcal{U}_{\mathcal{C}}(A)| \ge q^2 - k - \binom{k}{2}(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{C}}(A)}{m}. \tag{11}$$

If A is $\mathcal{C}$-complete, then

$$\boxed{q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{m}.} \tag{12}$$

Proof. Apply Corollary 3.2 with $\mathcal{H} = \mathcal{C}$, using $|V_{\mathcal{C}}(A)| = q^2 - k$ and $I_{\mathcal{H}}(A) = I_{\mathcal{C}}(A)$. If A is $\mathcal{C}$-complete, then $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$, and the first inequality rearranges to (12). □

The first correction in (12) accounts for unavoidable concurrence among secants; the last accounts for secant incidence spent on the exempt conic.

Corollary 4.2 (Even-size equality spectrum)**.** *Let $k \ge 6$ be even. If a $\mathcal{C}$-complete k-arc has zero defect, then*

$$q \in \left\{ k-2,\ \binom{k-1}{2},\ \binom{k-1}{2} + 1 \right\}.$$

In the first two cases every point of $\mathcal{C}$ has index $k/2$; in the third, exactly $k-1$ conic points have that index.

If q is also a power of two, then either $k = q+2$ or the sole remaining arithmetic candidate is

$$(q,k) = (4096, 92).$$

In particular, no zero-defect $\mathcal{C}$-complete eight- or twelve-arc exists in a Desarguesian plane, while a zero-defect ten-arc can occur only for $q = 8$ or $q = 37$. In fact, Theorem 5.3 excludes $q = 37$ and shows that the underlying arc in every ten-point equality pair has the *regular-hyperoval concurrence design. Such pairs exist in* $\mathrm{PG}(2,8)$*, but this statement does not classify either their projective realizations or the prescribed conics disjoint from them.*

Consequently, if $Q \geq 4$ *is an even prime power and* $d \geq 2$*, no* $\mathcal{C}$*-complete* $(Q+2)$*-arc in* $\mathrm{PG}(2,Q^d)\setminus\mathcal{C}$ *has zero defect.*

Proof. Put $Q = k-2$ and let $Z = \{x \notin A : r(x) = m\}$. Then $m = Q/2+1$ and, by Theorem 3.5, $|Z| = Q^2-1$. Put $s = |Z \cap \mathcal{C}|$. Every positive conic index equals m, so $I_{\mathcal{C}}(A) = ms$. Substitution in the zero-defect identity and factorization give

$$s = q+1-\frac{(q-Q)(2q-Q(Q+1))}{2}.$$

The elementary arc bound $k \leq q+2$ gives $q \geq Q$. Since $0 \leq s \leq q+1$, either $q = Q$ or $q \geq B := Q(Q+1)/2$. Write $q = B+t$ in the latter case. Then

$$s = q+1-t(q-Q).$$

For $t \geq 2$, already at $t=2$ the subtracted term exceeds $q+1$ by

$$B-2Q+1 = \frac{(Q-1)(Q-2)}{2} > 0,$$

and the excess increases with t. Hence $t \in \{0,1\}$. The displayed values of s at $q = Q, B, B+1$ are respectively $q+1, q+1, Q+1$.

Suppose now that $q = 2^n$. The case $q = B$ is impossible: since Q is even, the coprime factors Q and $Q+1$ could have product a power of two only if $Q+1 = 1$. In the case $q = B+1$, put $x = 2k-3$. Then

$$x^2+7 = 8q = 2^{n+3}.$$

Nagell's theorem [15], independently formalized in Lean by Banwait [7], gives the positive solutions

$$x \in \{1,3,5,11,181\}.$$

Here $k \geq 6$ gives $x = 2k-3 \geq 9$, excluding $1,3,5$, while evenness of k gives $x \equiv 1 \pmod 4$, excluding 11. Hence $x = 181$ and $(q,k) = (4096,92)$. The remaining case $q = Q$ is $k = q+2$. This argument supplies only a necessary parameter pair; it does not assert that the corresponding matching design has a rank-three realization.

For $k = 8,10,12$, the three candidate orders are respectively

$$\{6,21,22\}, \qquad \{8,36,37\}, \qquad \{10,55,56\}.$$

Since a Desarguesian plane has prime-power order, this proves the stated small-size consequences.

For the final assertion, $Q^d \geq Q^2 > B+1$ when $Q \geq 4$ and $d \geq 2$.

For the ten-point refinement, Theorem 3.5 produces a rank-three $\mathrm{MATCH}(10,5,1)$ design. Apply Theorem 5.3.

For existence over $\mathbb{F}_8 = \mathbb{F}_2(\alpha)$, where $\alpha^3+\alpha+1 = 0$, distinguish the constituent conic

$$\mathcal{D}:\ XZ = Y^2$$

from the prescribed conic, and denote its nucleus by $N_{\mathcal{D}} = (0,1,0)$. Let

$$A = \mathcal{D}\cup\{N_{\mathcal{D}}\}, \qquad \mathcal{C}:\ X^2+Y^2+Z^2+(1+\alpha)YZ = 0.$$

The gradient of the latter quadratic is $(0,(1+\alpha)Z,(1+\alpha)Y)$, so $\mathcal{C}$ is nonsingular. Its values on the eight affine points $(t^2,t,1)$ of A, in the polynomial-basis order $t=0,1,\alpha,1+\alpha,\dots,1+\alpha+\alpha^2$, are

$$1,\ \alpha,\ 1+\alpha^2,\ \alpha+\alpha^2,\ \alpha,\ 1,\ \alpha+\alpha^2,\ 1+\alpha^2,$$

and its value at each of the two infinite points of A is 1. Hence $A\cap\mathcal{C}=\varnothing$. Every point outside a hyperoval in $\mathrm{PG}(2,8)$ lies on five of its secants: the ten hyperoval points pair on the secants through that point. Thus every point of $\mathcal{C}$ has index five, so A is $\mathcal{C}$-complete and has zero defect. $\square$

Corollary 4.3 (Explicit integer-valued lower bound)**.** *Define*

$$L_1(q)=\min\left\{k\ge 3: q^2-k\le\binom{k}{2}(q-1)\right\},$$
$$L_2(q)=\min\left\{k\ge 3: q^2-k\le\binom{k}{2}(q-1)-\frac{6}{\lfloor k/2\rfloor}\binom{k}{4}\right\}.$$

Then

$$\rho_{\mathcal{C}}(q)\ge L_2(q)\ge L_1(q). \tag{13}$$

Proof. For $q\ge 3$ a set of at most two points has at most one secant, and that secant together with the set and the conic contains at most $2q+2<q^2+q+1$ points. Thus a $\mathcal{C}$-complete arc has at least three points, so the moment bound applies to the unrestricted minimum defining $\rho_{\mathcal{C}}(q)$. The definition of $L_2(q)$ involves only integer arithmetic after multiplication by $\lfloor k/2\rfloor$. $\square$

For calculation, the corrected capacity has the parity-dependent forms

$$\binom{k}{2}(q-1)-\frac{6}{\lfloor k/2\rfloor}\binom{k}{4}=\begin{cases}\dfrac{k-1}{2}\big(k(q-1)-(k-2)(k-3)\big), & k\text{ even},\\ \dfrac{k}{2}\big((k-1)(q-1)-(k-2)(k-3)\big), & k\text{ odd}.\end{cases} \tag{14}$$

For comparison, some values are

q	5	8	9	11	13	16	17	19
$L_1(q)$	4	5	5	6	6	7	7	7
$L_2(q)$	4	6	6	6	7	8	8	8

Thus the second-moment correction alone proves, for example, that a $\mathcal{C}$-complete arc in $\mathrm{PG}(2,16)$ has at least eight points.

The routine algebra behind the additive bound is isolated in the next lemma on the full interval needed to avoid any parity split.

Lemma 4.4 (Polynomial estimate)**.** *For $s\ge 2$ and $0\le a\le 2$, put*

$$P_s(a)=\frac{2a-3}{4}s^3+\frac{a^2-7a+10}{4}s^2+\frac{-3a^2+10a-8}{2}s+\frac{-a^3+5a^2-8a+6}{2}.$$

If $P_s(a)\ge 0$, then $a\ge 3/2-8/s$.

Proof. On $0 \le a \le 2$, the coefficients of s^2, s, and 1 in $P_s(a)$ are at most $5/2$, $1/6$, and 3, respectively; for the middle coefficient, $1/6 - (-3a^2 + 10a - 8)/2 = (3a - 5)^2/6$. Since $s \ge 2$, their total contribution is at most

$$\frac{5}{2}s^2 + \frac{1}{6}s + 3 \le 4s^2.$$

Thus $P_s(a) \ge 0$ implies $0 \le (2a - 3)s^3/4 + 4s^2$, and division by s^2 gives the claim. $\square$

Theorem 4.5 (Explicit additive lower bound)**.** *For every prime power q,*

$$\boxed{\rho_{\mathcal{C}}(q) \ge \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.} \tag{15}$$

Consequently, as q tends to infinity through prime powers,

$$\liminf_{q\to\infty} (\rho_{\mathcal{C}}(q) - \sqrt{2q}) \ge \frac{3}{2}. \tag{16}$$

Proof. For $q = 2$ the right side of (15) is negative, so the assertion is immediate. Assume $q \ge 3$. Let k be the size of a $\mathcal{C}$-complete arc and put $s = \sqrt{2q}$. The elementary first-moment inequality $q^2 - k \le \binom{k}{2}(q - 1)$ implies $k \ge s$ for $q \ge 2$: otherwise $k < s \le q$, while $k^2 < 2q$ gives $\binom{k}{2} < q$, so its right side is smaller than $q(q-1)$ whereas $q^2 - k \ge q(q-1)$. If $k \ge s+2$, then (15) is immediate. Otherwise write $k = s + a$, so $0 \le a < 2$ and $s \ge 2$.

Since $m = \lfloor k/2 \rfloor \le k/2$,

$$\frac{6}{m}\binom{k}{4} \ge \frac{12}{k}\binom{k}{4} = \frac{(k-1)(k-2)(k-3)}{2}.$$

Dropping the nonnegative conic term in (12) therefore gives the necessary inequality

$$q^2 - k \le \frac{k-1}{2}(k(q-1) - (k-2)(k-3)).$$

After substituting $q = s^2/2$ and $k = s + a$, the right side minus the left side is

$$\begin{aligned} &\frac{s+a-1}{2}\left((s+a)\left(\frac{s^2}{2} - 1\right) - (s+a-2)(s+a-3)\right) + (s+a) - \frac{s^4}{4} \\ &= \frac{2a-3}{4}s^3 + \frac{a^2 - 7a + 10}{4}s^2 + \frac{-3a^2 + 10a - 8}{2}s + \frac{-a^3 + 5a^2 - 8a + 6}{2} = P_s(a). \end{aligned}$$

It is nonnegative, so Lemma 4.4 gives $a \ge 3/2 - 8/s$, which is (15). Letting q tend to infinity through prime powers gives (16). $\square$

Remark 4.6 (The scale of the conic term). Every secant meets $\mathcal{C}$ in at most two points, so

$$0 \le I_{\mathcal{C}}(A) \le 2\binom{k}{2}.$$

At the relevant scale $k \asymp \sqrt{q}$, the term $I_{\mathcal{C}}(A)/m$ is therefore $O(\sqrt{q})$, while the universal overlap correction $6\binom{k}{4}/m$ is $\Theta(q^{3/2})$. Consequently, improving the lower bound on $I_{\mathcal{C}}(A)$ within (12) can be useful for finely balanced finite cases, but cannot improve the additive constant 3/2 in Theorem 4.5. Any stronger asymptotic result needs an additional mechanism, not merely a spectral estimate for $I_{\mathcal{C}}(A)$.

5 Equality classifications

The first two nontrivial orders illustrate the distinction between an abstract matching design and its rank-three projective realization.

Definition 5.1. A *rank-three projective realization* of a $\mathrm{MATCH}(k, m, 1)$ design $\mathcal{D}$ over a field K is an injective assignment $i \mapsto p_i \in \mathrm{PG}(2, K)$ whose image is a k-arc and such that, for every matching $M \in \mathcal{D}$, the m lines p_ip_j, $\{i, j\} \in M$, are concurrent. Two matching designs are *isomorphic* when a permutation of their k vertices carries the matching collection of one onto that of the other.

Corollary 5.2 (Six and seven points)**.** *(a) In a Desarguesian plane, if a six-arc has every intersection of two disjoint secants on a third secant, then the field has characteristic two, contains $\mathbb{F}_4$, and the arc is projectively equivalent to*

$$(1,0,0),\ (0,1,0),\ (0,0,1),\ (1,1,1),\ (1,\omega,\omega^2),\ (1,\omega^2,\omega), \qquad \omega^2+\omega+1=0.$$

Conversely, this configuration has the stated concurrence property. In particular, every zero-defect six-arc in a Desarguesian plane has this form.

(b) In any projective plane, no seven-arc has zero defect with respect to any prescribed hole set.

Proof. For six points, the three intersections of opposite sides of any complete quadrangle must all lie on the secant through the remaining pair. Normalize four vertices to

$$(1,0,0), \quad (0,1,0), \quad (0,0,1), \quad (1,1,1).$$

Their diagonal points are

$$(1,1,0), \qquad (1,0,1), \qquad (0,1,1).$$

Their determinant is -2, so they are collinear exactly in characteristic two, on $x+y+z=0$; that line contains the remaining pair. Write a fifth point as $(1, t, 1+t)$. Applying the diagonal-line condition to the quadrangle formed by the first three frame points and this fifth point gives the diagonal line

$$t(1+t)x + (1+t)y + tz = 0.$$

It must contain the remaining frame point $(1,1,1)$, which gives $t^2+t+1=0$; its intersection with $x+y+z=0$ is the sixth point $(1, t^2, t)$. Direct substitution verifies all required concurrences.

For seven points, Theorem 3.5 would produce a $\mathrm{MATCH}(7, 3, 1)$ design. Alspach and Heinrich [4, Theorem 3.1] proved that no such design exists. □

Theorem 5.3 (Rank-three ten-point classification)**.** *Let K be a field. Among the simple* $\mathrm{MATCH}(10, 5, 1)$ *designs, only the regular-hyperoval design can have a rank-three projective realization. It has one over K if and only if*

$$\operatorname{char} K = 2, \qquad \textit{and} \qquad \mathbb{F}_8 \subseteq K.$$

The other abstract design has no rank-three realization over any field.

Proof. Alspach and Heinrich allow repeated matchings in their definition, but when $\lambda = 1$ a repeated five-matching would repeat each of its pairs of edges and is therefore impossible. Thus their statement that Mathon found precisely two nonisomorphic $\mathrm{MATCH}(10, 5, 1)$ designs applies to Definition 5.1 [4, pp. 39–40]. Equivalently, these are the two partial geometries denoted $\mathrm{pg}(5, 7, 3)$ in Reichard and Woldar's (line size, point degree, α) convention, or $\mathrm{pg}(4, 6, 3)$ in the common $\mathrm{pg}(s, t, \alpha)$ convention. Reichard and Woldar identify Mathon's result as a full classification and construct models of both classes in Proposition 5.1.5 and Corollary 5.1.6 [18, Sec. 6]. This published two-class completeness is an external input. The paper-local enumeration in Appendix A reconstructs and checks representatives of the two classes, but is not used to prove that the list is complete.

Here is the realization test. Relabel and normalize four vertices to

$$p_0 = (1, 0, 0), \quad p_1 = (0, 1, 0), \quad p_2 = (0, 0, 1), \quad p_3 = (1, 1, 1).$$

No remaining point lies on p_0p_1, so write $p_i = (x_i, y_i, 1)$ for $4 \le i \le 9$. For each matching block, the first two secants determine its proposed concurrence point; requiring each of the other three secants to pass through that point gives three determinant equations. The 63 blocks therefore give 189 integral equations, and together with the arc inequations they are necessary and sufficient for a realization.

Only a necessary part of the arc open set is needed for the exclusions. Indeed,

$$\det(p_1, p_2, p_9) = x_9, \qquad \det(p_1, p_3, p_9) = x_9 - 1,$$

so every arc realization has $x_9(x_9 - 1) \ne 0$. The certificate saturates the concurrency ideal by this product. This enlarges, rather than shrinks, the locus that must be excluded: a unit saturated ideal therefore rules out every arc realization without claiming that this one product encodes all arc inequations.

Exact rational module identities make both saturated ideals the unit ideal after inverting 2; because their only denominator prime is 2, the same identities exclude both designs in every odd characteristic. In characteristic two the second design again has unit saturated ideal. For the regular-hyperoval design, the surviving triangular basis instead forces a parameter t with $t^3 + t + 1 = 0$; the arc inequations force $t \notin \mathbb{F}_2$, so $\mathbb{F}_8 \subseteq K$. The regular hyperoval over $\mathbb{F}_8$ and scalar extension prove the converse. Appendix A records the certificate, independent checks, and replay boundary. □

6 An upper-bound transfer from complete arcs

Let $t_2(2, q)$ denote the smallest size of an ordinary complete arc in $\mathrm{PG}(2, q)$.

Theorem 6.1 (Projective averaging). *Let $H \subseteq \mathrm{PG}(2, q)$ and let B be an ordinary complete b-arc. If*

$$b|H| < q^2 + q + 1, \tag{17}$$

then some projective image of B is disjoint from H and complete outside H.

Proof. Choose a uniformly random projectivity $g \in \mathrm{PGL}(3, q)$. Point transitivity gives

$$\mathbb{E}|gB \cap H| = \frac{b|H|}{q^2 + q + 1} < 1.$$

Since the intersection size is a nonnegative integer, it is zero for some g. Projectivities preserve ordinary completeness, so gB is complete outside H. □

Corollary 6.2 (Conic transfer)**.** *If $t_2(2,q) \le q$, then*

$$\rho_{\mathcal{C}}(q) \le t_2(2,q). \tag{18}$$

More generally, every complete b-arc with $b \le q$ has a projective image disjoint from the prescribed conic.

Proof. Take $H = \mathcal{C}$, so $|H| = q+1$. For integral $b \le q$ one has $b(q+1) < q^2+q+1$, and Theorem 6.1 applies. □

The hypothesis $t_2(2,q) \le q$ is part of the corollary, not a uniform assertion about every order. The application below uses only the range of sufficiently large q, where the Kim–Vu bound supplies it.

Kim and Vu proved that, for all sufficiently large q, every projective plane of order q has a complete arc of size at most $\sqrt{q}(\log q)^c$ for an absolute constant c [13]. This is less than q for large q, so Corollary 6.2 gives the following immediate consequence.

Corollary 6.3. *For some absolute constant c and all sufficiently large prime powers q,*

$$\rho_{\mathcal{C}}(q) \le \sqrt{q}(\log q)^c.$$

The transfer does not exploit the conic and is unlikely to be sharp. It nevertheless supplies a uniform upper bound without having to preserve the arc condition during a separate random covering construction.

7 Even characteristic and the nucleus

Assume q is even and let ν be the nucleus of $\mathcal{C}$. The nucleus is outside $\mathcal{C}$, all tangents to $\mathcal{C}$ pass through ν, and every line through ν is one of these $q+1$ tangents.

Proposition 7.1 (Case $\nu \in A$)**.** *Let A be a k-arc disjoint from $\mathcal{C}$ with $\nu \in A$. Exactly $k-1$ A-secants are tangent to $\mathcal{C}$. In particular,*

$$I_{\mathcal{C}}(A) \ge k-1, \qquad I_{\mathcal{C}}(A) \equiv k-1 \pmod 2.$$

If A is $\mathcal{C}$-complete, then

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{k-1}{m}. \tag{19}$$

Proof. For every $a \in A \setminus \{\nu\}$, the line νa is tangent to $\mathcal{C}$, giving $k-1$ tangent secants. No line joining two points of $A \setminus \{\nu\}$ can be tangent: every tangent contains ν, which would place three points of A on one line. Hence these are exactly the tangent A-secants.

Every tangent contributes one to $I_{\mathcal{C}}(A)$, while every other line contributes zero or two. This proves the lower bound and parity statement. Inequality (19) follows from Corollary 4.1. □

Proposition 7.2 (Case $\nu \notin A$)**.** *Let A be a $\mathcal{C}$-complete arc with $\nu \notin A$. Then*

$$1 \le r(\nu) \le m,$$

the tangent A-secants are exactly the secants through ν, and

$$I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2, \qquad I_{\mathcal{C}}(A) \ge r(\nu) \ge 1.$$

Consequently,

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}. \tag{20}$$

Proof. The nucleus is a required point outside $A \cup \mathcal{C}$, so relative completeness gives $r(\nu) \geq 1$; Lemma 2.1 gives the upper bound. A line is tangent to $\mathcal{C}$ exactly when it passes through ν. Thus the number of tangent A-secants is $r(\nu)$. Reducing the sum of the intersection sizes $|\ell \cap \mathcal{C}|$ modulo two gives $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2$. The remaining claims follow from Corollary 4.1. □

Corollary 7.3 (Universal even-characteristic loss)**.** *Every $\mathcal{C}$-complete arc in even characteristic satisfies*

$$I_{\mathcal{C}}(A) \geq 1.$$

Consequently, inequality (12) *can always be strengthened by replacing its final term by at least $-1/m$.*

Proof. If the nucleus belongs to A, Proposition 7.1 gives $I_{\mathcal{C}}(A) \geq k - 1 \geq 1$. Otherwise Proposition 7.2 gives $I_{\mathcal{C}}(A) \geq r(\nu) \geq 1$. □

These two cases are useful structural reductions, but Remark 4.6 limits what the incidence term alone can accomplish. For example, at $(q, k) = (16, 8)$ inequality (12) would be contradicted only by $I_{\mathcal{C}}(A) > 268$, whereas the universal upper bound is $I_{\mathcal{C}}(A) \leq 56$. Thus the exact determination of $\rho_{\mathcal{C}}(16)$ below requires information beyond a lower bound on $I_{\mathcal{C}}(A)$ inserted into the present inequality.

8 Verified finite-field examples

The verified values at $q = 5, 8, 9, 11$ use the lower bound $L_2(q)$ and explicit coordinates. At $q = 5$ the witness is the projective four-frame. At $q = 16$, the same ingredients give $8 \leq \rho_{\mathcal{C}}(16) \leq 9$; a checked projective classification excludes eight. The proof has three steps: the defect bound supplies the lower bound eight; a hypothetical $\mathcal{C}$-complete eight-arc would have its ordinary-uncovered locus on a quadratic avoiding the arc; and the certificate rules out every such quadratic, while the displayed nine-point witness supplies the upper bound.

8.1 Evaluation obstruction

The algebraic rejection used in that classification is an instance of the following elementary linear-algebra observation. Linear systems of quadrics containing arcs are an established tool; see Glynn [11] and Ball [6]. Fix nonzero vector representatives of the projective points under consideration. If V is a finite-dimensional linear system of homogeneous forms, write $\mathrm{ev}_x \in V^*$ for evaluation at the representative of x. The zero or nonzero status of this evaluation is independent of the chosen representative.

Indeed, if an arc is complete outside a conic, then its ordinary-uncovered locus lies in the zero set of the conic equation while every selected point avoids that zero set. The next lemma isolates exactly the linear-algebraic obstruction used to rule this out at $q = 16$.

Lemma 8.1 (Uncovered evaluation obstruction)**.** *Let A and U be finite sets of projective points. There is no nonzero $f \in V$ such that $U \subseteq Z(f)$ and $A \cap Z(f) = \varnothing$ if either of the following holds:*

1. *the evaluation map $f \mapsto (\mathrm{ev}_u(f))_{u \in U}$ is injective;*
2. *for some $a \in A$, the functional ev_a belongs to the span of $\{\mathrm{ev}_u : u \in U\}$.*

Proof. If f vanishes on U, injectivity in the first case gives $f = 0$. In the second case, applying a spanning relation for ev_a to f gives $f(a) = 0$, contrary to $A \cap Z(f) = \varnothing$. □

For the full system of degree-d forms, the evaluation functionals are the degree-d Veronese images of the points, up to nonzero scalar. Thus the first alternative says that the uncovered Veronese images span the dual coefficient space; the second supplies a selected Veronese image already in their span. In particular, for the full degree-d system it is enough to exhibit $\binom{d+2}{2}$ uncovered points whose Veronese evaluation matrix is invertible. This formulation is not specific to conics.

8.2 Small values and the classification over $\mathbb{F}_{16}$

Proposition 8.2. *For the standard conic*

$$\mathcal{C}: \; XZ = Y^2$$

in the indicated projective planes,

$$\rho_{\mathcal{C}}(5) = 4, \qquad \rho_{\mathcal{C}}(8) = 6, \qquad \rho_{\mathcal{C}}(9) = 6, \qquad \rho_{\mathcal{C}}(11) = 6.$$

Proof. Equation (13) gives

$$L_2(5) = 4, \qquad L_2(8) = L_2(9) = L_2(11) = 6.$$

For $q = 5$, the standard four-frame has ordinary-uncovered locus

$$V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

a nonsingular six-point conic. An invertible coordinate change carries this conic to $XZ = Y^2$ and the frame to the four coordinates in Appendix B; the relative certificate is checked in Lean. The appendix also lists $\mathcal{C}$-complete arcs of size six for $q = 8, 9, 11$. Their verification is described below. □

For context, the companion treatment classifies all conic-filling loci for $4 \le k \le 7$ [1, the conic-filling classification]. That broader classification is not used in Proposition 8.2.

The ordinary classification of eight-arcs over $\mathbb{F}_{16}$ records arc counts, stabilizers, secant-index distributions, and conic-contained classes. The theorem below adds the datum needed here: the ordinary uncovered locus either imposes six independent quadratic conditions or forces every containing quadratic to meet the arc.

The proof uses an exhaustive covering list, not pairwise inequivalent representatives. The following contract isolates the mathematical completeness step; Appendix A records the normalized point encoding, generator, certificate checker, and trust boundary. Let F denote the standard projective four-frame.

Lemma 8.3 (Augmentation certificate contract)**.** *For $4 \le i \le 8$, let $\mathcal{L}_i$ be a finite list of i-arcs with $\mathcal{L}_4 = \{F\}$. Suppose that for every $4 \le i < 8$, every $P \in \mathcal{L}_i$, and every $x \notin P$ such that $P \cup \{x\}$ is an arc, the certificate supplies a child $P' \in \mathcal{L}_{i+1}$ and an invertible linear map whose induced projectivity carries $P \cup \{x\}$ onto P'. Then every eight-arc in* $\mathrm{PG}(2, 16)$ *is projectively equivalent to a member of $\mathcal{L}_8$.*

Proof. Choose four points of the eight-arc. Every four-arc is a projective frame, so a projectivity carries those points to F. Insert the remaining four points in any order. At each step the stated certificate transports the legal insertion to the next list. Induction through the four layers gives a member of $\mathcal{L}_8$ projectively equivalent to the original arc. □

Theorem 8.4 (Quadratic avoidance over $\mathbb{F}_{16}$)**.** *Let A be any eight-arc in $\mathrm{PG}(2,16)$, and let $U(A)$ be the points outside A that lie on no secant of A. There is no nonzero homogeneous quadratic form Q such that*

$$U(A) \subseteq Z(Q) \qquad \text{and} \qquad A \cap Z(Q) = \varnothing.$$

The assertion includes singular quadratic forms.

Proof. Covering list. The checked list lengths are $|\mathcal{L}_5| = 4$, $|\mathcal{L}_6| = 61$, $|\mathcal{L}_7| = 454$, and $|\mathcal{L}_8| = 2633$. Every legal insertion among all 273 projective points has a checked child and projective transporter, so Lemma 8.3 proves exhaustiveness. The final length agrees with the independent class count of Al-Seraji–Al-Ogali [3, Theorem 3.8.1], but that agreement is not used.

Leaf obstruction. For 2630 leaves, six uncovered points have an invertible quadratic evaluation matrix. For each of the other three, evaluation at an arc point lies in the span of evaluations on the uncovered locus. These are the two alternatives of Lemma 8.1. Projective transport preserves the assertion, so the checked leaf result holds for every eight-arc. ☐

Corollary 8.5 (Exact relative value over $\mathbb{F}_{16}$)**.** *For every nonsingular conic $\mathcal{C} \subseteq \mathrm{PG}(2,16)$,*

$$\rho_{\mathcal{C}}(16) = 9.$$

Proof. Equation (13) gives $L_2(16) = 8$. If a $\mathcal{C}$-complete eight-arc existed, its ordinary-uncovered locus would lie in the zero set of the nonzero quadratic defining $\mathcal{C}$, while the arc is disjoint from that zero set. This contradicts Theorem 8.4. The nine-point witness in Appendix B gives the matching upper bound. ☐

9 Uncovered-locus reconstruction

This coda does not apply to the $\mathcal{C}$-complete arcs studied above: they satisfy $U(A) \subseteq \mathcal{C}$, and in the square-root range $k < q$ the first-moment covering inequality gives $\binom{k}{2} \geq q + 1$, opposite to the strict hypothesis below.

For fixed arc length and sufficiently large q, the ordinary-uncovered locus remembers its parent arc. The inverse map has two elementary stages. Put $N = \binom{k}{2}$ and $S = \mathrm{PG}(2,q) \setminus U(A)$. First recover the secant arrangement from the gap between $q + 1$, the number of points on a genuine secant, and N, the largest number of points in which a nonsecant line can meet the union. Then recover the vertices from the gap between their secant multiplicity $k - 1$ and the largest possible nonvertex multiplicity $\lfloor k/2 \rfloor$:

$$U(A) \longmapsto S \longmapsto \{\ell : |\ell \cap S| > N\} \longmapsto \{x : \deg(x) = k - 1\} = A.$$

The strict first gap is the only large-field hypothesis.

Proposition 9.1 (Uncovered-locus reconstruction)**.** *Let A and B be k-arcs in $\mathrm{PG}(2,q)$, where $k \geq 3$ and*

$$q + 1 > \binom{k}{2}.$$

If $U(A) = U(B)$ as literal subsets of the plane, then $A = B$ as unlabelled point sets.

Proof. The complement of $U(A)$ is the union of the $\binom{k}{2}$ distinct secants of A, and likewise for B. Let ℓ be a secant of A. If no secant of B were equal to ℓ, then the secants of B would cover at most $\binom{k}{2}$ points of ℓ, one per line. This is impossible because $|\ell| = q + 1 > \binom{k}{2}$ and the two secant unions are equal. Thus every secant of A is a secant of B; since the two sets have the same finite size, their secant-line sets coincide.

It remains to recover the vertices from this line arrangement. Each point of A lies on its $k-1$ incident secants. A point outside A lies on at most $\lfloor k/2 \rfloor$ secants: distinct secants through it use disjoint pairs of the k vertices and hence form a matching. Since $k-1 > \lfloor k/2 \rfloor$ for $k \geq 3$, the set A is exactly the set of points incident with $k-1$ recovered secants, and the same description recovers B. □

Corollary 9.2 (Canonical inverse and automorphisms)**.** *Under the hypotheses of Proposition 9.1, put $S = \mathrm{PG}(2,q) \setminus U(A)$ and $N = \binom{k}{2}$. The secants of A are exactly the lines ℓ with $|\ell \cap S| > N$, and A is exactly the set of points incident with $k-1$ such lines. Moreover,*

$$\mathrm{Stab}_{\mathrm{P\Gamma L}_3(q)}(U(A)) = \mathrm{Stab}_{\mathrm{P\Gamma L}_3(q)}(A).$$

Proof. Every secant lies in S and has $q+1 > N$ points. A nonsecant line meets each of the N secants in at most one point, so it contains at most N points of their union S. The vertex-recovery rule is the one proved above. Finally, every semilinear projectivity stabilizing A stabilizes $U(A)$. Conversely, if $gU(A) = U(A)$, then $U(gA) = gU(A) = U(A)$, so Proposition 9.1 gives $gA = A$. □

Remark 9.3 (Sharpness of the strict threshold). The strict inequality cannot in general be weakened to $q+1 \geq \binom{k}{2}$. For the projective four-frame A over $\mathbb{F}_5$, Proposition 8.2 gives $U(A) = \mathcal{C}(\mathbb{F}_5)$, while $q+1 = \binom{4}{2} = 6$. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}_3(5)$ has order $|\mathrm{PGL}_2(5)| = 120$, whereas the setwise stabilizer of a projective frame has order $4! = 24$. The conic stabilizer therefore moves A through at least five distinct four-frames, all with the same literal uncovered locus $\mathcal{C}(\mathbb{F}_5)$.

10 Conclusion and further questions

The prescribed-hole identity retains the local equality and stability data discarded by the usual scalar secant count. At zero defect those local conditions assemble into a matching design, while for a conic they sharpen the integer lower bound and isolate the finite obstruction at $q = 16$. Projective averaging supplies the complementary upper transfer. The reconstruction coda shows that, above the sharp incidence threshold, the complete uncovered locus also remembers the secant arrangement and its vertices.

Problem 10.1. Construct $\mathcal{C}$-complete arcs of size $O(\sqrt{q})$, or prove that this is impossible for an infinite family. Theorem 6.1 currently supplies only $O(\sqrt{q}(\log q)^c)$. On the lower-bound side, Remark 4.6 shows that improving the additive constant $3/2$ requires a mechanism coupling rank-three realization to the matching structure, rather than a sharper estimate of the conic-incidence term.

Problem 10.2. Develop the higher-dimensional analogue for caps complete outside a prescribed quadric in $\mathrm{PG}(n,q)$. The evaluation lemma already applies to arbitrary feature maps and Veronese degrees; the missing ingredient is an appropriate exact secant-moment remainder for caps.

A Computational and formal verification

The body separates mathematical reductions from finite exclusion. The corresponding evidence and trust boundaries are summarized below. In the source supplement, `lean/`

`RelativeConicArcs/TRUST.md` records exact commands, toolchain versions, SHA-256 digests, theorem names, and frozen outputs. The import-only gate `lean/RelativeConicArcs/Gates/Relconic.lean` is the kernel-checking entry point for precisely the results used here. The pinned versions are Lean 4.32.0-rc1, Lake `5.0.0-src+b4812ae`, and Mathlib revision `571b8a8e54219b4d393f75f4b8653fac08197fcc`. From the supplement's `lean/` directory, the exact replay is:

```
nix develop --command lake exe cache get
nix develop --command env LEAN_NUM_THREADS=1 \
  lake build RelativeConicArcs.Gates.Relconic
```

Success is exit status zero with a final Lake success report for the named gate; the gate itself has no runtime output. With fetched dependencies, a direct single-thread elaboration took 26 seconds on the development host. A fresh replay should allow 30 minutes, 8 GiB of RAM, and 10 GiB of free disk. The largest generated $q = 16$ leaves have a measured peak of approximately 1.3 GiB each.

The complete `#print axioms` output for the gate theorems consists only of `propext`, `Classical.choice`, and `Quot.sound`. The checked sources contain no `sorry`, custom axiom declaration, or `native_decide`.

Claim	Certificate and checker	Remaining trust
Defect, bounds, transfer, and reconstruction	`Gates.Relconic`; named declarations in the trust manifest	Lean kernel, Mathlib, and `propext`, `Classical.choice`, and `Quot.sound`
$q = 16$ exclusion	augmentation transitions, $2630 + 3$ leaf records, and `Q16QuadraticTransport`, imported by the gate	Lean kernel and Mathlib; generator labels and deduplication are not trusted
Ten-point realization	`check_match10_rank_three.py` and its JSON certificate	Mathon's external two-class classification; Python integer arithmetic and Singular exact arithmetic
Ramanujan–Nagell classification	Nagell's proof [15]; independent Lean formalization by Banwait [7]	Nagell's published argument; for the formal cross-check, the Lean kernel and Mathlib
Small-field witnesses	`ExampleChecks/Q5.lean` through `Q16.lean`	Lean kernel and the explicit finite-field implementations

The ten-point bundle consists of `check_match10_rank_three.py`, its JSON certificate, and `check_match10_rank_three.sha256` in the paper directory. Running

```
nix shell nixpkgs#singular --command \
  python3 check_match10_rank_three.py --check
```

regenerates and checks the certificate. Singular Gröbner bases and module lifts remain trusted executions. A second monomial order and a direct $\mathbb{F}_8$ incidence construction provide independent checks; the same record checks the regular hyperoval, the disjoint prescribed conic used in Corollary 4.2, and its nonvanishing gradient. Abstract two-class completeness remains the cited external theorem. The odd-characteristic computation saturates by $x_9(x_9 - 1)$, the product of the two necessary arc determinants displayed in the proof of Theorem 5.3, and the checked rational module lifts have no denominator prime other than 2; hence their unit relations survive in every odd characteristic.

Lean also proves, without an extra hypothesis, the operational asymptotic statement along unbounded orders: every constant below 3/2 is eventually a lower bound for $\rho_{\mathcal{C}}(q) -$

$\sqrt{2q}$. The separate real-valued `liminf` wrapper assumes coboundedness only because its codomain $\mathbb{R}$ has no value $+\infty$; this assumption is not used in the paper's elementary passage to (16).

For the $q = 16$ exclusion, the supplement contains the source generator `papers/arcs_complete_outside_conic/search_rhoc16.cpp`, its frozen report `search_rhoc16_output.txt`, and their SHA-256 digests. The nine-line report is only a generation summary, not the certificate: the proof objects are the emitted Lean transition and leaf modules consumed by the gate. Each certified transition carries an invertible matrix and checked source–target scalar equalities. The Lean checker separately verifies that these transitions cover every legal augmentation of the standard four-frame. At the leaves it checks either six independent quadratic evaluation conditions or an explicit linear combination forcing a containing quadratic to meet the arc. Thus the theorem trusts the Lean kernel and the stated classical inputs, but neither the generator's canonical labels nor an external claim of exhaustive execution.

B Finite-field witnesses

All coordinates below are relative to the standard conic

$$\mathcal{C}: \; XZ = Y^2.$$

For $q = 8$ and $q = 16$, an integer is the binary coefficient vector of the polynomial basis $1, \alpha, \ldots$, with $\alpha^3 + \alpha + 1 = 0$ and $\alpha^4 + \alpha + 1 = 0$, respectively. For $q = 9$, the integer $a_0 + 3a_1$ denotes $a_0 + a_1\alpha$, where $\alpha^2 + 1 = 0$. Coordinates for $q = 5, 11$ are read modulo the prime. The checked witnesses are

q	A
5	$\{(1,0,3),(2,1,2),(3,3,4),(1,4,4)\}$
8	$\{(0,1,1),(0,1,2),(1,0,1),(1,0,2),(1,1,2),(1,1,6)\}$
9	$\{(1,0,4),(1,0,5),(1,1,0),(1,1,2),(1,2,3),(1,2,4)\}$
11	$\{(1,10,0),(1,9,1),(1,4,7),(1,8,5),(0,1,4),(1,1,7)\}$
16	$\{(1,5,14),(1,5,3),(1,9,2),(1,0,9),(1,13,13),$ $(1,6,10),(1,3,9),(0,1,11),(1,10,2)\}$

The files `lean/RelativeConicArcs/ExampleChecks/Q5.lean` through `Q16.lean` check disjointness from $\mathcal{C}$, the arc condition, and relative coverage. The independent verifier `papers/arcs_complete_outside_conic/verify_relative_conic_arcs.py` regenerates the same coverage data; its digest and frozen output are recorded in the trust manifest.

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